F Broader Impacts

This work focuses on understanding and controlling self-reflection in language models, which carries several potential societal implications. On the positive side, enabling more precise control over the reflection mechanisms in language models can lead to more reliable automated reasoning in high-stakes domains such as medical diagnosis, scientific research, and educational applications. By improving reasoning accuracy through reflection enhancement while maintaining computational efficiency, our technique could make high-quality AI assistance more accessible across resource-constrained environments.

However, we also recognize potential negative impacts. Enhanced reasoning capabilities could be misused for generating more convincing misinformation or deceptive content. Additionally, while our method aims to improve model efficiency, it could potentially increase the environmental impact of AI systems by encouraging more computation-intensive applications that rely on repeated reflection-enhanced reasoning.

Our approach requires direct access to model weights and hidden states, which represents a safeguard against casual misuse, as most harmful applications would be limited to those with substantial technical expertise and computational resources. Furthermore, we aim to mitigate risks by openly sharing our findings with the research community and focusing on techniques that can be used to make models more efficient rather than simply more powerful.

The experiments conducted in this research were limited to benchmark evaluations using publicly available datasets and did not involve human subjects or require IRB approval. Our work did not collect any new data from individuals, nor did it involve deployment in real-world settings that could directly impact people.

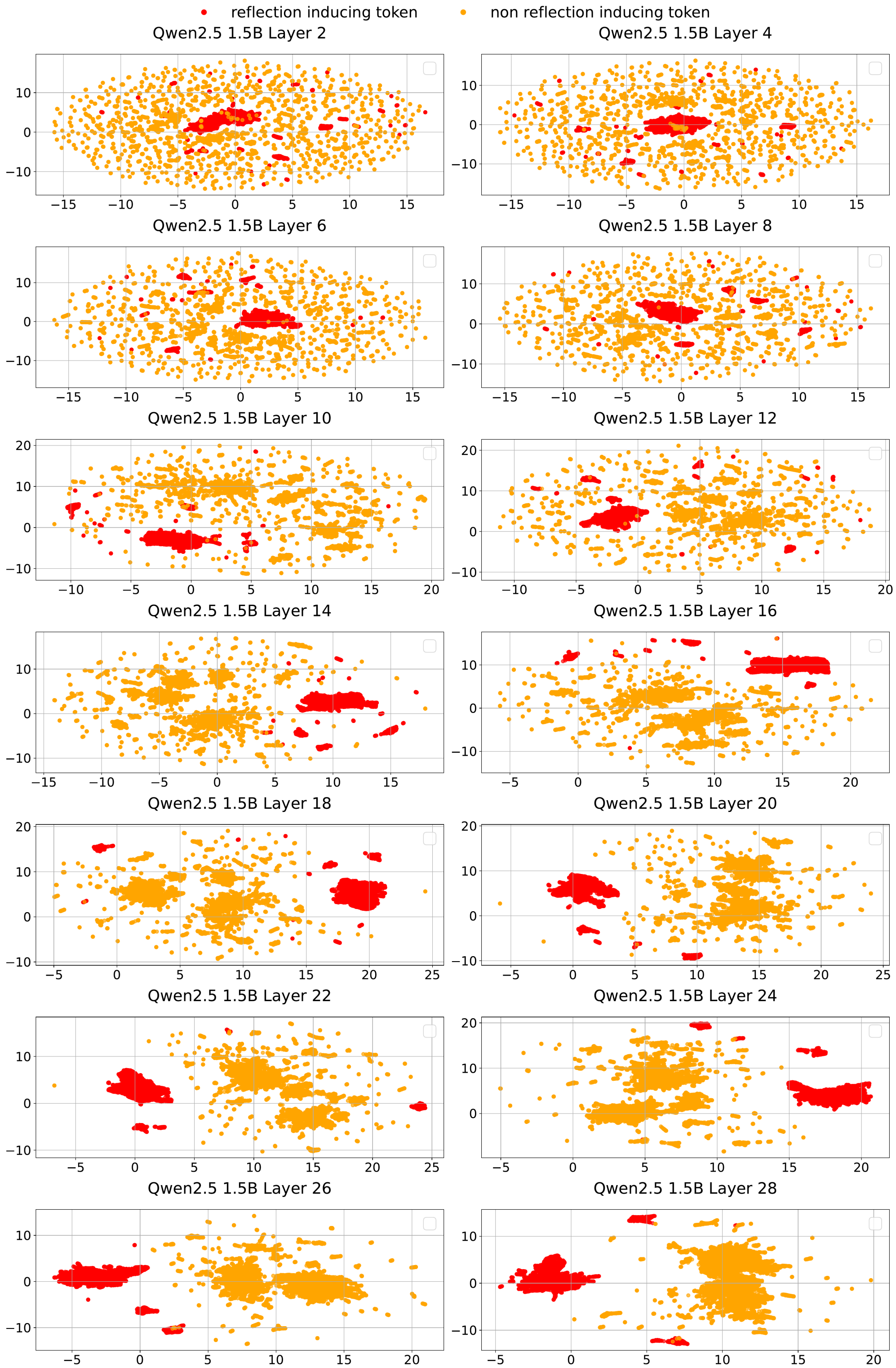

Figure 8: UMAP visualization of hidden states from the Qwen2.5-1.5B model across 14 layers.

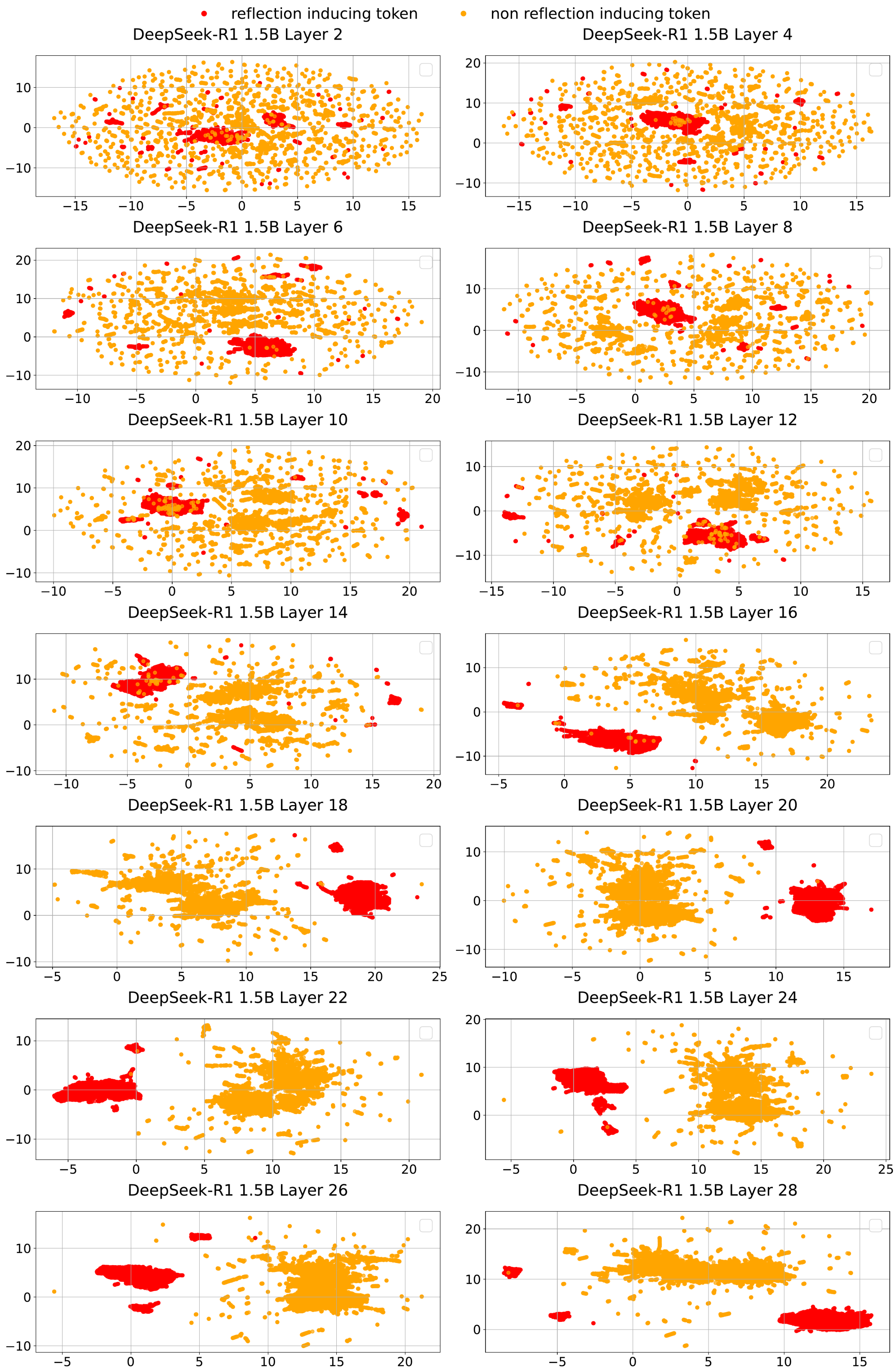

Figure 9: UMAP visualization of hidden states from the DeepSeek-R1-Distill-Qwen-1.5B model across 14 layers.